

Tableau Project

Project 1 Dashboard

Part 1: Profits & losses

First, we'll try to identify the important centers of profit and loss for the superstore.

1. Among the pairs of dimensions (e.g., subcategory + region, or shipping mode + product ID) which are the two biggest profit centers and two biggest loss-makers? Justify your conclusion with a visualization.

Sub-category/Region

Profit : Binders(77,309) / West(195,544)

Loss : Binders(-59,251) / Central(-78,120)

Shipping Mode / Product ID

Profit : Standard Class (325,168) / TEC-CO-100004722 (8,400)

Loss : Standard Class(-137,269) / TEC-MA-100000418(-6,600)

Between the pairs of dimension; The West Region (195,544) and Standard Class Shipping (325,168) are the biggest profit centers, while the Central Region (-78,120) and Standard Class Shipping (-137,296) are the biggest loss-makers.

2. Which products should the superstore stop selling? Justify your conclusion with a visualization.

The store should stop selling the Cubify CubeX 3D Printer Double Head Print, Cubify CubeX 3D Printer Triple Head Print, Lexmark MX611dhe Monochrome Laser Printer, GBC DocuBind P400 Electric Binding System, and the Ibico EPK-21 Electric Binding System.

These products have either turned out as loss-makers, or don't sell enough units to make a good profit.

(See "Loss-makers" sheet)

3. Which product subcategories should the store focus on and which should they stop selling? Choose 3 of each.

The store should focus on the Binders, Copiers, and Phones subcategories. They should stop selling Tables, Bookcases, and Supplies. (See "Sub-Category/Region" sheet)

Part 2: Advertising

The superstore wants to know if advertising would be worth it. Advertising works over time and geography, and your average profit per unit sold should be high enough to justify it.

1. Identify the 3 best combinations of states and month of the year to advertise in. Make a visualization of the average profit for each month of the year for those 3 states and argue how much you would be willing to pay in advertising for those states in those months.

You should be most willing to pay for advertisements based on the return on ad spend ratio. Let's say you should be willing to spend 1/5 of profits on advertising for this exercise.

AVG ROAS (Revenue / Ad Investment)

Indiana	Mar : $30 / 6 = 5$	Oct : $643 / 129 = 4.99$	Nov : $71 / 14 = 5$
Vermont	Mar : NULL	Oct : NULL	Nov : $596 / 119 = 5$
WA	Mar : $559 / 112 = 4.99$	Oct : $28 / 6 = 4.67$	Nov : $44 / 9 = 4.89$

I would be willing to pay for advertisements for Indiana in October (\$129), Vermont in November (\$119), and Washington in March (\$112). Considering that these are profits from unadvertised sales, I believe that the resulting profits would be even better than those projected through my calculations.

Part 3: Returned items

In the last part, we'll use the Returns table to see if some products have abnormal rates of being returned to the store.

HINT: Make sure the Returns table is LEFT JOIN'ed onto the Orders table. You should see both "Yes" and "null" values in the Returned column.

1. Make the Returned field into a calculated field where the null values are 0 and the Yes values are 1
2. Use this new field to make a visualization for each of the following questions:
 - Which products have the highest return rate?
Machines with a 35% return rate.
 - Which customers have the highest return rate?
Customers from the Western Region, with a 41% return rate.
3. In a separate sheet, make a visualization of the average profit against the average return rate on a dimension of your choice (state, shipping mode, product type, etc.). Present a visual argument why the superstore should do away with or keep doing business on the basis of this dimension.

The store should do away with data based on Postal codes. This dimension could be effective in making observations for a business that is only available in certain States or Regions. It's overwhelming to look at this visualization when the business has expanded across multiple Regions.